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Standard of care options — approved and guideline-endorsed strategies

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Standard of care options

The treatment strategies that are standard for this patient's situation, meaning a regulator approved them for a population that includes this patient or a major academic or clinical-society guideline carries them. This report runs alongside the targetable-feature ranking rather than in place of it, and it does not narrow what that ranking surfaces.

Assessed: 17 standard options. **Consider now:** 8. **Behind an open gate:** 2. **Already received:** 0.

Seventeen standard-of-care options were assessed for this systemic-therapy-naive metastatic uterine leiomyosarcoma; eight are actionable now. The guideline-preferred systemic choice at this decision point is single-agent doxorubicin, graded I,A by BSG 2025 and on the US label for metastatic soft tissue sarcoma. The planned doxorubicin plus trabectedin combination outperformed it in LMS-04 but is off-label first line and ranked on the Experimental table, so both regimens, plus gemcitabine and docetaxel as the fallback if cardiac or pulmonary screening fails, are competing for one chemotherapy-naive slot. The other consider-now rows are the supportive scaffolding an anthracycline demands and this record lacks: baseline echocardiogram with cumulative-dose planning, primary G-CSF prophylaxis against a 28% febrile neutropenia rate, fertility and ovarian-function counselling whose window closes at cycle 1, early palliative care, and tumour NGS with germline TP53 testing.

Nothing standard has been spent yet. The 2024 hysterectomy was surgery for a presumed fibroid, not therapy, so every systemic line remains available; the sequencing risk runs the other way, since trabectedin maintenance would spend the on-label second-line trabectedin slot, and dacarbazine would close a PARP-trial door.

Two rows sit behind gates: the pembrolizumab and TRK-inhibitor tumour-agnostic approvals, untested rather than negative, both closed or opened by the NGS panel already ordered. Lung-directed consolidation waits on the 2024 operative report and on whether induction converts the anatomy. Later lines are honestly thin: eribulin's LMS subgroup is null and pazopanib's uterine evidence is a 44-patient retrospective pooling.

This report adds the standard options alongside the case's targeted ranking; it does not narrow or reorder the options ranked elsewhere on this page.

Options to consider now

Standard options this patient is eligible for, with an endorsement that covers their situation, and no unresolved gate in front of them.

Doxorubicin, single agent

Priority: essential **Modality:** Systemic chemotherapy **Line:** First line **Intent:** life-prolonging intent

What makes it standard.

- other_society British Sarcoma Group (BSG): I,A, 'Standard first-line treatment is single-agent doxorubicin' (covers this patient · BSG UK guidelines 2025 (Br J Cancer))
 - covers: Advanced or metastatic soft tissue sarcoma, first-line systemic therapy.
- FDA on-label approval (covers this patient · DailyMed label accessed 2026-08-12)
 - covers: Doxorubicin is indicated for the treatment of metastatic soft tissue sarcoma (label section 1.2,

among other cancers).

Fit to this patient. Conditionally eligible She is systemic-therapy naive, so the first-line slot is open, but no baseline echocardiogram or LVEF exists before an anthracycline, no laboratory values were supplied at intake, and the ECOG of 1 is an intake assumption never documented by a clinician.

- blocker: No baseline echocardiogram or LVEF on file before an anthracycline
- blocker: ECOG 1 is an intake assumption, not clinician-documented
- blocker: No laboratory values of any kind were supplied at intake

Single-agent doxorubicin holds the strongest guideline position in this disease: BSG 2025 grades it I,A as standard first-line treatment, and metastatic soft tissue sarcoma is on the US label. It occupies the same first-line slot as the planned doxorubicin plus trabectedin combination, which beat it on PFS and OS in LMS-04 but remains off-label at this line and is ranked on the Experimental table. Whichever is chosen, the missing baseline echocardiogram is the true gate on cycle 1.

Key evidence.

- LMS-04 doxorubicin-alone arm (first-line LMS benchmark) · randomized phase 3 comparator arm · n=150 · median PFS 6.2 months and median OS 24 months in the doxorubicin-alone arm; grade 3-4 neutropenia 13%, febrile neutropenia 9%
- GeDDiS doxorubicin arm · randomized phase 3 comparator arm · n=257 · 46.3% alive and progression-free at 24 weeks; median PFS 23.3 weeks on doxorubicin

Toxicities that would change the decision. Cumulative, dose-capped cardiotoxicity; at her age the lifetime anthracycline ledger matters as much as this treatment course; Myelosuppression that must be read against altitude-shifted baseline blood counts at 8,500 ft; Febrile neutropenia 9% as a single agent in LMS-04, well below the combination's 28%

Alongside the targeted options. Competes for the same line of therapy. The guideline-preferred single agent and the planned off-label LMS-04 combination spend the same chemotherapy-naive first-line slot, and the same echocardiogram gates both. This row names the trade the treating team is already weighing; it does not resolve it.

Last verified. 2026-08-12

References. pmid:38734790; pmid:35835135; pmid:39231341; pmid:28882536; nct:NCT02997358

Baseline echocardiogram, cardiac surveillance and cumulative anthracycline planning

Priority: essential **Modality:** Supportive care **Line:** Not line-indexed **Intent:** risk reduction

What makes it standard.

- ASCO ASCO clinical practice guideline recommendation (covers this patient · ASCO 2017 (J Clin Oncol))
 - covers: Prevention and monitoring of cardiac dysfunction in adults receiving potentially cardiotoxic therapy, including anthracyclines: discuss risk before beginning therapy, with prevention and screening strategies during treatment for higher-risk patients.
- other_society 2022 ESC cardio-oncology guideline (developed with EHA, ESTRO and IC-OS) (covers this patient · ESC 2022 (Eur Heart J))
 - covers: Baseline cardiovascular risk assessment including echocardiography before anthracycline

chemotherapy, with surveillance during and after treatment.

Fit to this patient. Eligible Nothing in the profile blocks an echocardiogram; the problem is that it has not been done and induction is imminent.

A baseline echocardiogram before an anthracycline is standard care under both the ASCO 2017 and ESC 2022 cardio-oncology guidelines, and it was an eligibility criterion of the trial her planned regimen comes from; none exists in her chart. At her age the cumulative doxorubicin dose is a lifetime ledger, so the dose-cap and dexrazoxane conversation belongs before cycle 1, noting honestly that the dexrazoxane label population (metastatic breast cancer after 300 mg/m² of doxorubicin) is not hers. A pathogenic germline TP53 result would re-open the whole genotoxic plan.

Alongside the targeted options. Can run alongside the targeted options. The echocardiogram in this row is the same one the Experimental table's rank-1 workup names as a true gate on cycle 1 of the ranked induction regimen; a failed screen moves the first-line conversation to this table's gemcitabine plus docetaxel row.

Last verified. 2026-08-12

References. pmid:27918725; pmid:36017568; nct:NCT02997358

Fertility and ovarian-function counselling before gonadotoxic chemotherapy

Priority: essential **Modality:** Supportive care **Line:** Not line-indexed **Intent:** risk reduction

What makes it standard.

- ASCO ASCO clinical practice guideline recommendation (covers this patient · ASCO 2018 update (J Clin Oncol))
 - covers: Providers should initiate the discussion of possible infertility with patients treated during their reproductive years as early as possible before treatment starts; oocyte cryopreservation is standard practice; evidence for GnRH-agonist ovarian suppression is conflicting and it should not replace proven methods.

Fit to this patient. Eligible She is in her thirties with presumably retained ovaries after a hysterectomy done for a presumed fibroid; the profile records no counselling and no reproductive-endocrinology referral.

She is 30-39 and days from chemotherapy that can end ovarian function, and the record shows no one has raised it; hysterectomy settled childbearing anatomy but not oocytes, hormones or menopause timing. ASCO places this discussion before therapy starts, as early as possible, because the widest option set exists only before cycle 1. What ovarian function and the hormonal aftermath mean to her is one of the four questions the advocate flagged as never asked.

Alongside the targeted options. Comes before the targeted options. Any oocyte retrieval or suppression decision has to land before the ranked induction begins. This names the timing pressure; it is not an argument to delay treatment, and the window is days rather than weeks.

Last verified. 2026-08-12

References. pmid:29620997

Primary G-CSF prophylaxis through induction chemotherapy

Priority: essential **Modality:** Supportive care **Line:** Not line-indexed **Intent:** risk reduction

What makes it standard.

- ASCO ASCO guideline recommendation (covers this patient · ASCO 2015 WBC growth factors update)
 - covers: Prophylactic use of colony-stimulating factors is warranted when the risk of febrile neutropenia is approximately 20% or higher and no equally effective and safe regimen that does not require CSFs is available.

Fit to this patient. Eligible No contraindication in the profile; the planned combination regimen's 28% febrile neutropenia rate puts her above the ASCO prophylaxis threshold from cycle 1.

The planned induction produced febrile neutropenia in 28% of trial patients, above the roughly 20% threshold at which ASCO recommends primary G-CSF prophylaxis, so growth-factor support belongs in the cycle 1 orders rather than being discovered at cycle 2. Neutropenic-fever counselling also needs the altitude context: baseline counts and oximetry at 8,500 ft shift the reference ranges the fever thresholds are read against.

Key evidence.

- LMS-04 combination-arm cytopenia rates · randomized phase 3 safety read · n=150 · grade 3-4 neutropenia 80% and febrile neutropenia 28% in the doxorubicin plus trabectedin arm, vs 13% and 9% with doxorubicin alone

Toxicities that would change the decision. Bone pain is the common G-CSF toxicity; the trade is against a 28% febrile neutropenia rate

Alongside the targeted options. Can run alongside the targeted options. Belongs in the cycle 1 orders of the ranked induction regimen. If single-agent doxorubicin is chosen instead, its 9% febrile neutropenia rate sits below the ASCO threshold, so the G-CSF call follows the regimen choice.

Last verified. 2026-08-12

References. pmid:26169616; pmid:35835135

Tumour NGS with germline TP53 testing before genotoxic chemotherapy

Priority: essential **Modality:** Other **Line:** Not line-indexed **Intent:** risk reduction

What makes it standard.

- ESMO ESMO Precision Medicine Working Group recommendation, 2024 update (covers this patient · ESMO PMWG 2024 (Ann Oncol))
 - covers: Tumour NGS recommendations expanded to encompass rare tumours including sarcoma, and tumour NGS is recommended to detect tumour-agnostic alterations in metastatic cancers where access to matched therapies is available.
- other_society ERN GENTURIS guideline recommendation (covers this patient · ERN GENTURIS 2020 (Eur J Hum Genet))
 - covers: Li-Fraumeni and heritable TP53-related cancer syndromes: soft-tissue sarcoma sits in the tumour

spectrum, and radiotherapy and conventional genotoxic chemotherapy contribute to subsequent primary tumours in carriers, so germline status is best established before such therapy.

Fit to this patient. Eligible The 2024 resection block exists and no result is pending; only the consent conversation and a records and block-release request stand between her and the panel.

Tumour NGS in sarcoma moved from clinician preference to guideline-backed workup in the ESMO 2024 precision-medicine update, and ERN GENTURIS puts the germline TP53 answer before genotoxic chemotherapy because uLMS sits in the Li-Fraumeni spectrum. On this table the panel is the key that opens or closes both tumour-agnostic approval doors. The germline draw is a consent conversation with consequences for her relatives, and whether she wants that information has not been asked.

Alongside the targeted options. Can run alongside the targeted options. This is the same panel, consent conversation and block-allocation question the Experimental table carries at rank 1, and it is what opens or closes the pembrolizumab and TRK rows below. One requisition set serves both tables.

Last verified. 2026-08-12

References. pmid:38834388; pmid:32457520

Gemcitabine + docetaxel

Priority: high **Modality:** Systemic chemotherapy **Line:** First line **Intent:** life-prolonging intent

What makes it standard.

- other_society British Sarcoma Group (BSG): II,B (covers this patient · BSG UK guidelines 2025 (Br J Cancer))
 - covers: First-line alternative to doxorubicin across sarcoma subtypes: GeDDiS showed the regimen non-inferior to doxorubicin but more toxic; activity demonstrated in leiomyosarcoma and undifferentiated pleomorphic sarcoma.

Fit to this patient. Conditionally eligible Chemotherapy-naive, so the slot is open; docetaxel needs a hepatic panel that does not exist yet, and pneumonitis surveillance needs the pulmonary baseline and altitude-adjusted oximetry the rank-1 workup orders.

- blocker: No hepatic panel on file (docetaxel dosing)
- blocker: No pulmonary function tests or oximetry baseline at her 8,500 ft residence for pneumonitis surveillance
- blocker: ECOG 1 is an intake assumption, not clinician-documented

The main first-line alternative when an anthracycline is unwise: BSG 2025 carries it at II,B as non-inferior to doxorubicin though more toxic, and it has dedicated first-line uLMS data (ORR 35.8% in GOG 087L). The PI's synthesis names it the fallback if the cardiac or pulmonary baseline disqualifies doxorubicin, and that is exactly where it sits on this table. Its pneumonitis liability deserves naming in advance against her lungs and her altitude.

Key evidence.

- GeDDiS · randomized phase 3, first line, all STS subtypes · n=257 · 46.4% vs 46.3% alive and progression-free at 24 weeks; PFS HR 1.28 (95% CI 0.99-1.65) not favouring gemcitabine plus docetaxel; more toxic than doxorubicin
- GOG 087L · single-arm phase 2, first-line metastatic uLMS · n=42 · ORR 35.8% (95% CI 23.5-49.6); median

PFS 4.4 months

Toxicities that would change the decision. Both drugs carry pneumonitis risk, which would be hard to read against bilateral infiltrative lung disease and altitude-shifted oximetry; Docetaxel fluid retention, neuropathy and myelosuppression; GeDDiS logged more toxicity than doxorubicin

Alongside the targeted options. Competes for the same line of therapy. Holds the fallback slot for the same first-line window the ranked LMS-04 regimen occupies: a failed echocardiogram or pulmonary screen moves the first-line conversation here without touching the rest of the ranking.

Last verified. 2026-08-12

References. pmid:38734790; pmid:28882536; pmid:18534250; pmid:12065559

Early specialist palliative care alongside active treatment

Priority: high **Modality:** Supportive care **Line:** Not line-indexed **Intent:** symptom palliation

What makes it standard.

- ASCO ASCO guideline recommendation (covers this patient · ASCO 2024 update (J Clin Oncol))
 - covers: Patients with advanced solid tumours should be referred to specialised interdisciplinary palliative care teams beginning early in the disease course, alongside active treatment of their cancer.
- other_society British Sarcoma Group (BSG) recommendation (covers this patient · BSG UK guidelines 2025 (Br J Cancer))
 - covers: Supportive and palliative care should always be considered in advanced disease, with early referral to specialist palliative care services.

Fit to this patient. Eligible Metastatic disease with cough and dyspnea; nothing blocks a referral, and none is recorded.

Concurrent specialist palliative care from early in metastatic disease is an ASCO recommendation rather than a late-stage handoff, and BSG 2025 says the same for advanced sarcoma. In this case it also hosts the conversation nobody has had: goals of care were never discussed at intake and every preference on file is an assumed default. A symptom-control referral is a natural place for the four unasked questions to finally get asked.

Alongside the targeted options. Can run alongside the targeted options. Runs alongside induction and every later branch; it neither delays nor competes with any ranked option.

Last verified. 2026-08-12

References. pmid:38748941; pmid:38734790

Doxorubicin + ifosfamide

Priority: medium **Modality:** Systemic chemotherapy **Line:** First line **Intent:** disease-control intent

What makes it standard.

- other_society British Sarcoma Group (BSG): II,B (covers this patient · BSG UK guidelines 2025 (Br J Cancer))

- covers: First-line advanced STS: the combination has not been shown to improve survival over single-agent doxorubicin; response rates and PFS are higher at greater toxicity, and it may be considered in individual patients where a response would improve symptoms.

Fit to this patient. Conditionally eligible Her age band fits the EORTC 62012 population (18-60, PS 0-1), but the missing baselines bite harder here: ifosfamide needs the renal panel that does not exist, on top of the absent echocardiogram.

- blocker: No renal panel on file (ifosfamide nephrotoxicity)
- blocker: No baseline echocardiogram before an anthracycline
- blocker: ECOG 1 is an intake assumption, not clinician-documented

Carried at II,B for the situation where tumour shrinkage itself is the goal, since the combination buys response rate and PFS but not survival (EORTC 62012 OS HR 0.83 with the upper bound crossing 1). Two case-specific facts keep it at a completeness listing: leiomyosarcoma is among the less ifosfamide-sensitive histologies, and her symptomatic lung disease already has a response-seeking plan in the ranked induction.

Key evidence.

- EORTC 62012 · randomized phase 3, OS primary endpoint · n=455 · median OS 14.3 vs 12.8 months, HR 0.83 (95.5% CI 0.67-1.03, p=0.076); higher ORR and PFS with substantially more toxicity

Toxicities that would change the decision. Febrile neutropenia 46% vs 13% with doxorubicin alone in EORTC 62012; Ifosfamide encephalopathy and nephrotoxicity; Shares the cumulative anthracycline ledger with every other doxorubicin-containing option

Alongside the targeted options. Competes for the same line of therapy. A third occupant of the single chemotherapy-naïve first-line slot the ranked LMS-04 regimen is planned to spend; listing it does not reopen the choice, it maps the slot.

Last verified. 2026-08-12

References. pmid:24618336; pmid:38734790

Options at the next decision point

Standard for a line this patient has not reached yet, or for a disease state that has not arrived. Listed now so the sequence is visible in advance.

Pazopanib

Priority: high **Modality:** Targeted therapy **Line:** Second line **Intent:** disease-control intent

What makes it standard.

- FDA on-label approval (partial population match · FDA approval 2012-04; DailyMed label accessed 2026-08-12)
 - covers: Adults with advanced soft tissue sarcoma who have received prior chemotherapy; efficacy not demonstrated in adipocytic STS or GIST.
- other_society British Sarcoma Group (BSG): II,B (partial population match · BSG UK guidelines 2025 (Br J

Cancer))

- covers: Metastatic STS (not liposarcoma) after prior chemotherapy, on a 3-month progression-free survival improvement over placebo.

Fit to this patient. Conditionally eligible Label and trial evidence both require prior chemotherapy, and she is systemic-therapy naive until induction runs; the population match becomes exact at first progression.

- blocker: Label requires prior chemotherapy; she has had no systemic therapy yet

The on-label second-line anchor: approved for advanced STS after prior chemotherapy on PALETTE's PFS effect (HR 0.31), carried by BSG 2025 at II,B. Its uterine-LMS-specific evidence is thinner than the label reads: a retrospective pooling of 44 heavily pretreated uterine sarcoma patients with ORR 11% and median PFS 3.0 months. It becomes live at first progression, where it shares the slot with the HRD-conditional PARP branch and the phase 1 doors the Experimental table ranks.

Key evidence.

- PALETTE · randomized double-blind placebo-controlled phase 3 · n=369 · median PFS 4.6 vs 1.6 months, HR 0.31 (95% CI 0.24-0.40); overall survival not significantly improved
- EORTC uterine sarcoma retrospective (Benson) · retrospective pooling of two prospective EORTC trials · n=44 · ORR 11% (95% CI 3.8-24.6); median PFS 3.0 months

Toxicities that would change the decision. Hypertension and transaminitis with scheduled liver-function monitoring; Pneumothorax has been reported in sarcoma patients with lung metastases; VEGFR-class wound-healing impairment would interact with any later lung-directed surgery

Alongside the targeted options. Competes for the same line of therapy. At first progression this approved anchor and the ranked experimental doors ask for the same treatment slot; an on-label option a payer recognises exists at that line whichever way the team goes. Its wound-healing liability would also matter if metastasectomy ever became real.

Last verified. 2026-08-12

References. pmid:22595799; pmid:27012429; pmid:38734790

Trabectedin monotherapy

Priority: medium **Modality:** Systemic chemotherapy **Line:** Second line **Intent:** disease-control intent

What makes it standard.

- FDA on-label approval (partial population match · FDA approval 2015-10; DailyMed label accessed 2026-08-12)
 - covers: Adults with unresectable or metastatic liposarcoma or leiomyosarcoma who received a prior anthracycline-containing regimen.
- other_society British Sarcoma Group (BSG): licensed second-line agent (partial population match · BSG UK guidelines 2025 (Br J Cancer))
 - covers: Licensed as second-line treatment for all soft tissue sarcomas, with activity in leiomyosarcoma and liposarcoma; NICE approves continuation until disease progression.

Fit to this patient. Conditionally eligible The label requires a prior anthracycline she has not yet had; more

importantly, the planned first line already contains trabectedin through maintenance, so at progression the drug would usually count as spent.

- blocker: Label requires a prior anthracycline-containing regimen; she is systemic-therapy naive
- blocker: The planned LMS-04 first line delivers trabectedin to progression through its maintenance phase, which would close this on-label second-line use

On label this is the classic second-line drug for uLMS after an anthracycline, with a randomized PFS advantage over dacarbazine (HR 0.55) that held in the uterine subset. Her case bends the sequence: the planned first line already delivers trabectedin to progression through its maintenance phase, which would spend this slot before she reaches it. If the first line ends up being doxorubicin alone instead, this row returns to the table intact.

Key evidence.

- ET743-SAR-3007 (trabectedin vs dacarbazine) · randomized phase 3 · n=518 · median PFS 4.2 vs 1.5 months, HR 0.55 (95% CI 0.44-0.70); grade 3+ ALT elevation 25%, treatment-related deaths 2.1%
- uLMS subset of the same trial · subset analysis · n=232 · PFS HR 0.57, mirroring the parent trial

Toxicities that would change the decision. Grade 3+ transaminitis in a quarter of patients; hepatic panel and CK stay on a recurring schedule; Rhabdomyolysis 1.2% and treatment-related mortality 2.1% in the registration trial

Alongside the targeted options. May be foreclosed by a targeted option. Progression on LMS-04 maintenance is progression on trabectedin, which closes the on-label second-line use of the same drug; single-agent doxorubicin first line would leave it open. The upstream regimen choice decides whether this row survives, and that is a sequencing fact rather than advice.

Last verified. 2026-08-12

References. pmid:26371143; pmid:28651804; pmid:38734790

Gemcitabine + dacarbazine (or dacarbazine alone)

Priority: medium **Modality:** Systemic chemotherapy **Line:** Third line or later **Intent:** disease-control intent

What makes it standard.

- other_society British Sarcoma Group (BSG): II,B (covers this patient · BSG UK guidelines 2025 (Br J Cancer))
 - covers: Gemcitabine plus dacarbazine as an alternative later-line regimen in advanced STS, generally well tolerated.

Fit to this patient. Likely eligible No case-specific block once earlier lines are spent; no organ-function labs exist yet, as everywhere in this record.

A tolerable later-line option with a randomized signal: gemcitabine plus dacarbazine beat dacarbazine alone on PFS and OS in previously treated STS, and BSG 2025 carries it at II,B. Its main case-specific weight is a foreclosure: dacarbazine exposure would exclude her from the randomized trial behind the Experimental table's PARP branch, so sequencing, not efficacy, is what this row is really about.

Key evidence.

- Spanish sarcoma group randomized phase 2 (Garcia-Del-Muro) · randomized phase 2 vs dacarbazine alone ·

n=113 · median PFS 4.2 vs 2.0 months (HR 0.58, 95% CI 0.39-0.86); median OS 16.8 vs 8.2 months (HR 0.56, 95% CI 0.36-0.90)

Toxicities that would change the decision. Generally well tolerated; granulocytopenia the main serious event, febrile neutropenia uncommon; Gemcitabine pneumonitis risk reads poorly against her infiltrative lung disease

Alongside the targeted options. May foreclose a targeted option. The randomized successor trial behind the ranked PARP branch excludes prior temozolomide or dacarbazine, so giving this regimen first would close that trial door while the HRR/HRD result that would make the PARP branch live is still unmeasured. The order in which this row is reached matters more than usual.

Last verified. 2026-08-12

References. pmid:21606430; pmid:38734790; nct:NCT05432791

Pulmonary metastasectomy

Priority: medium **Modality:** Surgery **Line:** Not line-indexed **Intent:** disease-control intent

What makes it standard.

- other_society British Sarcoma Group (BSG): III,B (partial population match · BSG UK guidelines 2025 (Br J Cancer))
 - covers: Selected patients with thoracic metastases: long-term survivors are reported (20-40% of patients undergoing lung surgery) although prospective data are few and the survival benefit remains unproven.

Fit to this patient. Likely ineligible Bilateral infiltrative, possibly inflammatory pulmonary disease is the anatomy the surgical series screened out, and the staging record is incomplete: the 2024 operative report has not been retrieved.

- blocker: Bilateral infiltrative disease pattern; the resection series selected patients with countable, resectable lesions
- blocker: Morcellation status of the 2024 resection unknown; if morcellation occurred, peritoneal restaging precedes any consolidation planning (abdominopelvic dissemination 44% vs 12.9% in the cited series)
- blocker: No pulmonary function tests or oximetry baseline, read against an 8,500 ft residence

Guideline-carried for selected patients, and the long-term survivors in the registries are real, but every series is retrospective and operability-selected, and bilateral infiltrative disease is exactly what they screened out. The honest framing is the PI's: a question to re-ask if induction converts the anatomy, not a survival number to quote today. Two records gate it before anatomy even matters: the 2024 operative report and a measured pulmonary baseline at her altitude.

Key evidence.

- International Registry of Lung Metastases · retrospective multi-institutional registry, operability-selected · n=5206 · 5-year OS 36% after complete resection; median OS 35 vs 15 months for complete vs incomplete resection
- Brigham LMS metastasectomy series · retrospective single-institution cohort · n=82 · median OS 70 months in the LMS subset vs 24 months non-LMS (p=0.049)

Toxicities that would change the decision. Thoracotomy morbidity and permanently lost pulmonary reserve, priced differently for someone living at 8,500 ft

Alongside the targeted options. Comes after the targeted options. Only becomes assessable if the ranked induction converts bilateral infiltrative disease into a resectable pattern, and after the rank-1 workup retrieves the 2024 operative report; the morcellation answer decides whether peritoneal restaging comes first.

Last verified. 2026-08-12

References. pmid:38734790; pmid:9011700; pmid:21867989; pmid:21565389

Stereotactic body radiotherapy (SBRT) to pulmonary metastases

Priority: medium **Modality:** Radiotherapy **Line:** Not line-indexed **Intent:** disease-control intent

What makes it standard.

- other_society British Sarcoma Group (BSG): named option, evidence limited (partial population match · BSG UK guidelines 2025 (Br J Cancer))
 - covers: Stereotactic ablative radiotherapy has become another potential option for lung metastases from STS, with evidence still limited.

Fit to this patient. Likely ineligible The SBRT series treated small numbers of countable lesions with a controlled primary; bilateral diffuse infiltrative disease is not currently a stereotactic target, and no pulmonary baseline exists.

- blocker: Bilateral diffuse infiltrative disease is not an oligometastatic SBRT target; the cited series treated up to a few discrete lesions
- blocker: No pulmonary function baseline at 8,500 ft; lost lung volume carries a higher price at altitude
- blocker: Consolidation intent shares the unresolved 2024 morcellation and staging question with the surgical row

The non-surgical consolidation route, named by BSG 2025 as a potential option with limited evidence. The series are retrospective and eligibility-selected, and their headline numbers measure different things: the same 44-patient cohort shows 90% local control at 24 months and 46% survival at 24 months, so ablating visible lesions and changing the disease course are separate claims. Like metastasectomy, this is a question to re-ask if induction converts bilateral infiltrative disease into countable targets.

Key evidence.

- MGH/Penn sarcoma lung SBRT · retrospective two-institution cohort · n=44 · local control 90% (95% CI 84-96) at 24 months, alongside overall survival 46% at 24 months in the same patients
- Humanitas sarcoma lung SBRT · retrospective single-institution cohort · n=28 · actuarial 5-year local control 96%

Toxicities that would change the decision. Radiation pneumonitis would be difficult to separate from disease progression and altitude physiology in these lungs

Alongside the targeted options. Comes after the targeted options. Like the surgical row, this is a consolidation question that only opens if the ranked induction converts the anatomy; the same operative-report and staging gates apply.

Last verified. 2026-08-12

References. pmid:38734790; pmid:32588468; pmid:25686482

Eribulin

Priority: low **Modality:** Systemic chemotherapy **Line:** Third line or later **Intent:** disease-control intent

What makes it standard.

- FDA on-label approval (liposarcoma only) (different population · FDA approval 2016-01; DailyMed label accessed 2026-08-12)
 - covers: Unresectable or metastatic liposarcoma after a prior anthracycline-containing regimen; leiomyosarcoma is not in the US label.
- other_society British Sarcoma Group (BSG): noted, liposarcoma authorisation (different population · BSG UK guidelines 2025 (Br J Cancer))
 - covers: Marketing authorisation for unresectable liposarcoma; data in other sarcoma types limited; not currently funded in England.

Fit to this patient. Conditionally eligible Medically she could receive it after an anthracycline, but the label covers liposarcoma, so any use in her histology is off-label and rests on a null subgroup.

- blocker: Requires a prior anthracycline-containing regimen; not yet given
- blocker: US label excludes leiomyosarcoma, so use in her histology is off-label

Carried for completeness and honesty rather than promise: the OS benefit that won eribulin its approval came from the liposarcoma stratum, the US label is liposarcoma-only, and the leiomyosarcoma subgroup is null (OS HR 0.93 with PFS numerically worse than dacarbazine). For her histology this is a drug the later-line conversation will mention and the data do not support well. That thin later-line shelf is itself a finding of this screen.

Key evidence.

- Study 309 · randomized open-label phase 3 vs dacarbazine · n=452 · median OS 13.5 vs 11.5 months, HR 0.77 (95% CI 0.62-0.95), driven by the liposarcoma stratum
- Study 309 leiomyosarcoma subgroup · preplanned subgroup analysis · n=309 · OS HR 0.93 (95% CI 0.71-1.20, p=0.57); median PFS 2.2 vs 2.6 months, numerically worse than dacarbazine

Toxicities that would change the decision. Neutropenia and neuropathy, the familiar microtubule-inhibitor profile

Alongside the targeted options. Competes for the same line of therapy. By third line the approved shelf in LMS is thin, which is the backdrop against which the ranked phase 1 doors were surfaced; this row shares that slot without arguing against them.

Last verified. 2026-08-12

References. pmid:26874885; pmid:31065111; pmid:38734790

Standard options behind an open gate

Endorsed for this patient's situation, but gated on a result that is missing, pending, or below threshold. The gate has to close before the option is real.

Pembrolizumab (tumour-agnostic: MSI-H/dMMR or TMB-H)

Priority: low **Modality:** Immunotherapy **Line:** Second line **Intent:** disease-control intent

What makes it standard.

- FDA tumour-agnostic approval (MSI-H/dMMR) (population match unclear · FDA accelerated approval 2017-05; DailyMed label accessed 2026-08-12)
 - covers: Adult and pediatric patients with unresectable or metastatic MSI-H or dMMR solid tumours, as determined by an FDA-authorized test, that have progressed following prior treatment and who have no satisfactory alternative treatment options.
- FDA tumour-agnostic approval (TMB-H) (population match unclear · FDA accelerated approval 2020-06; DailyMed label accessed 2026-08-12)
 - covers: Adult and pediatric patients with unresectable or metastatic TMB-high (10 or more mutations per megabase) solid tumours, as determined by an FDA-authorized test, that have progressed following prior treatment and who have no satisfactory alternative treatment options.

Fit to this patient. Not determinable from the record Both gating biomarkers are untested rather than negative, and the label also requires progression on prior therapy, a line she has not yet reached. The missing facts are the MSI/MMR and TMB results the ordered NGS panel will return.

Biomarker gate. MSI-H/dMMR or TMB MSI-H or dMMR by an FDA-authorized test; or TMB ≥ 10 mut/Mb (never measured)

Two tumour-agnostic approval doors that testing would open: pembrolizumab is approved for MSI-H/dMMR and for TMB-H solid tumours after progression on prior therapy. Both markers are currently untested rather than negative, both are rare in uLMS, and both close cheaply on the NGS panel already in the rank-1 workup. Until a result returns, the population match is unknowable, which is why this row sits behind a workup gate rather than in the consider-now group.

Key evidence.

- KEYNOTE-158 (MSI-H/dMMR noncolorectal) · single-arm phase 2, 27 tumour types · n=233 · ORR 34.3% (95% CI 28.3-40.8); median PFS 4.1 months
- KEYNOTE-158 TMB analysis · prospectively planned retrospective analysis · n=790 · ORR 29% (95% CI 21-39) in TMB ≥ 10 mut/Mb vs 6% (95% CI 5-8) below

Toxicities that would change the decision. Immune-related pneumonitis would be hard to read against bilateral infiltrative lung disease and altitude-shifted oximetry

Alongside the targeted options. Can run alongside the targeted options. The gate closes on the same one-pass NGS panel the Experimental rank-1 workup orders; nothing extra is spent. A positive would move this row to on-label care at progression.

Last verified. 2026-08-12

References. pmid:31682550; pmid:32919526

TRK inhibitor (larotrectinib or entrectinib), tumour-agnostic if NTRK fusion-positive

Priority: low **Modality:** Targeted therapy **Line:** Any line **Intent:** disease-control intent

What makes it standard.

- FDA tumour-agnostic accelerated approval (population match unclear · FDA accelerated approval 2018-11; Vitrakvi label revised 4/2025, DailyMed accessed 2026-08-12)
 - covers: Adult and pediatric patients with solid tumours that have an NTRK gene fusion without a known acquired resistance mutation, are metastatic or where resection would cause severe morbidity, and have no satisfactory alternative treatments or have progressed following treatment.

Fit to this patient. Not determinable from the record NTRK fusion status has never been tested; the profile is silent on the single fact that gates the entire option.

Biomarker gate. NTRK gene fusion NTRK1/2/3 fusion without a known acquired resistance mutation, ideally by RNA-inclusive fusion calling (never measured)

The second tumour-agnostic door: TRK inhibitors carry a pooled ORR of 79% in NTRK fusion-positive tumours, but the fusion is vanishingly rare in uLMS and hers is untested rather than negative. The requisition detail decides the answer's quality, since several common panels are DNA-only and the rank-1 workup already specifies RNA-inclusive fusion calling. A positive would be on-label care; the prior is low.

Key evidence.

- Larotrectinib pooled analysis · pooled analysis of three phase 1/2 trials · n=159 · ORR 79% (95% CI 72-85)

Toxicities that would change the decision. Dizziness, fatigue and weight gain; a benign profile relative to the cytotoxic rows

Alongside the targeted options. Can run alongside the targeted options. Priced by the same rank-1 requisition; the workup already specifies RNA-inclusive fusion calling because a DNA-only panel could return a false negative that reads as a closed question.

Last verified. 2026-08-12

References. pmid:32105622

Assessed and set aside

Screened against this patient and found not to apply. Listed for the audit trail.

Aromatase inhibition (letrozole)

Priority: low **Modality:** Endocrine therapy **Line:** Any line **Intent:** disease-control intent

What makes it standard.

- other_society British Sarcoma Group (BSG): option limited to receptor-expressing tumours (different population · BSG UK guidelines 2025 (Br J Cancer))

- covers: Endocrine therapy in uterine sarcoma is reserved for hormone-receptor-expressing, typically more indolent tumours; the supporting phase 2 required ER and/or PR expression.

Fit to this patient. Likely ineligible ER <1% and PR <1% by IHC, recorded as a confirmed decision-relevant negative; the trial that supports letrozole in uLMS required receptor expression and saw durable benefit only above 90% staining.

- blocker: ER <1% and PR <1% by IHC, a confirmed decision-relevant negative
- blocker: Specimen provenance of the ER/PR stain (2024 primary vs 2026 lung core) is unverified; the foreclosure holds only as firmly as that check

Listed so its absence is explained rather than wondered about: endocrine therapy in uLMS belongs to receptor-expressing tumours, and hers stained under 1% for both ER and PR, which closes this door. The one residual thread is provenance: if the stain came off the 2024 block rather than current disease, the restain in the rank-1 workup could in principle reopen the question, though a true conversion is unlikely.

Key evidence.

- Letrozole phase 2 in ER/PR-expressing uLMS · single-arm phase 2 · n=27 · 12-week PFS 50% (90% CI 30-67); durable benefit confined to tumours staining above 90%

Last verified. 2026-08-12

References. pmid:38734790; pmid:24222211